



Pharm*Achieve*

PHARMACY MANAGEMENT MODULE 1: COMMUNITY PHARMACY

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

Providing Care:
Clinical Care

Providing Care:
Distribution

Knowledge
& Expertise

Communication
& Collaboration

Leadership &
Stewardship

Professionalism

TABLE OF CONTENTS

1 Pharmacy Formats

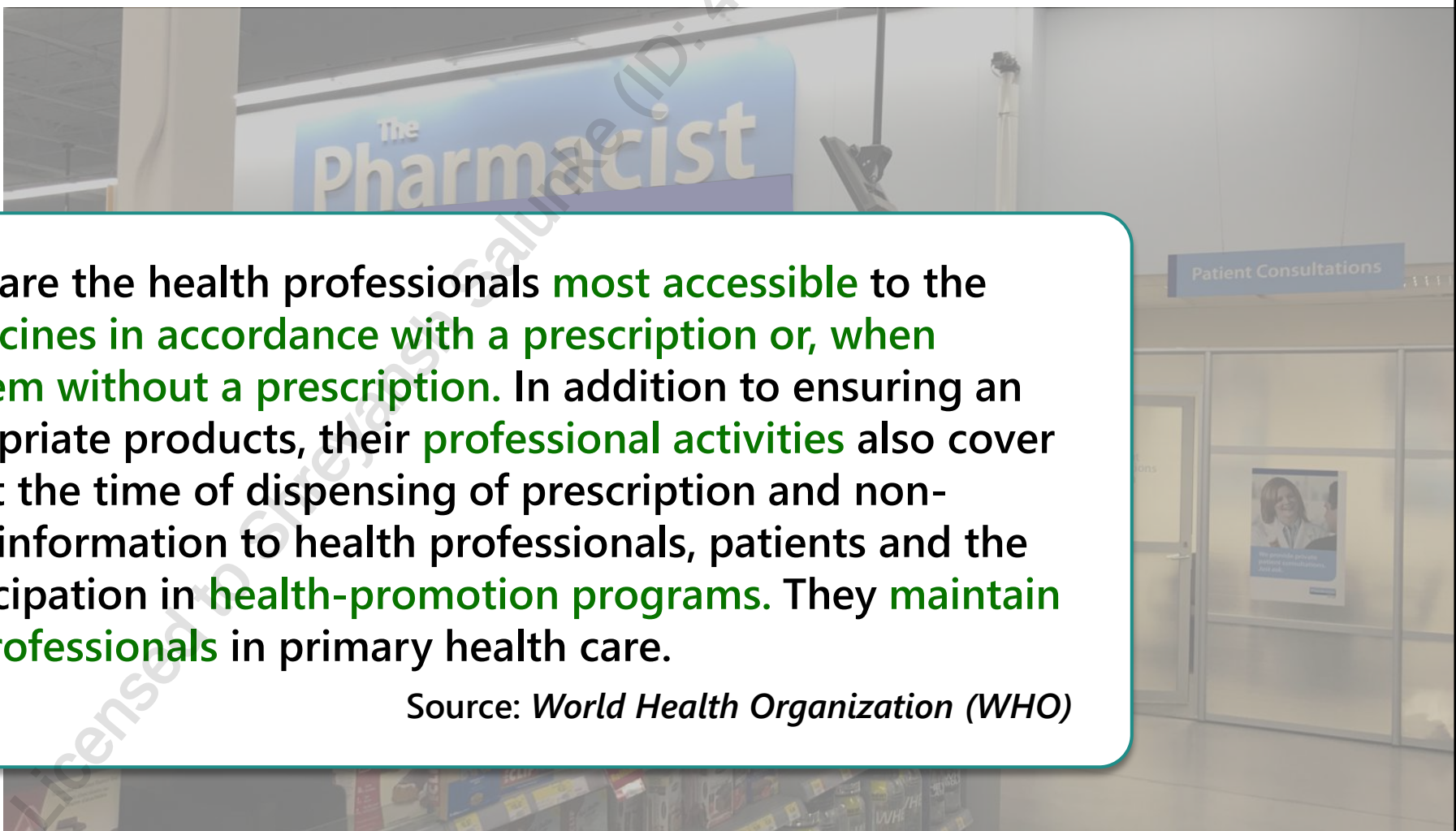
2 Business Options and the Canadian Supply Chain

3 Product Acquisition

4 Storage

5 Dispensing Process

UNDERSTANDING COMMUNITY PRACTICE



Community pharmacists are the health professionals **most accessible** to the public. **They supply medicines in accordance with a prescription or, when legally permitted, sell them without a prescription.** In addition to ensuring an accurate supply of appropriate products, their **professional activities** also cover **counselling** of patients at the time of dispensing of prescription and non-prescription drugs; drug information to health professionals, patients and the general public; and participation in **health-promotion programs**. They **maintain links with other health professionals** in primary health care.

Source: *World Health Organization (WHO)*

PHARMACY FORMATS IN CANADA

Independent	Chain	Franchise	Co-operative/Banner
- Owned by a licensed pharmacist - Provides flexibility to tailor services to meet the needs of their patients (Ex: niche services such as specialty compounding, travel vaccinations, veterinary medicine or home health care) - Each province or territory has different requirements to own a pharmacy, however, most require the pharmacy to be <u>owned in whole or majority</u> by a pharmacist licensed in the regulated jurisdiction	- Typically national - Can be a stand-alone pharmacy (e.g. Rexall) or be located inside a mass merchandiser store (e.g. Loblaws, Costco, Sobeys, Walmart) - Corporately owned, operating decisions are made by their head office	- A franchise combines independent pharmacy and chain into one (E.g. Medicine Shoppe also Shoppers Drug Mart) - Owned by a licensed pharmacist who pays a fee for use of the banner name and the right to sell products with franchise branding	- Independent pharmacy, but like franchise (e.g. Pharmasave, IDA) - Owner joins the banner, have a purchasing share in the company (owner becomes shareholder) - Owner can keep shares in the company if they are owner of the store

PHARMACY FORMATS IN CANADA

Tele-pharmacy

- Allows dispensing of medication and providing patient counselling to remote communities that may not have access to a traditional pharmacy
- There is an **accredited traditional pharmacy and a remote location**; these are in different locations and will communicate using a secure videoconferencing platform (for example, Zoom, WebEx, Microsoft Teams)

Central Fill

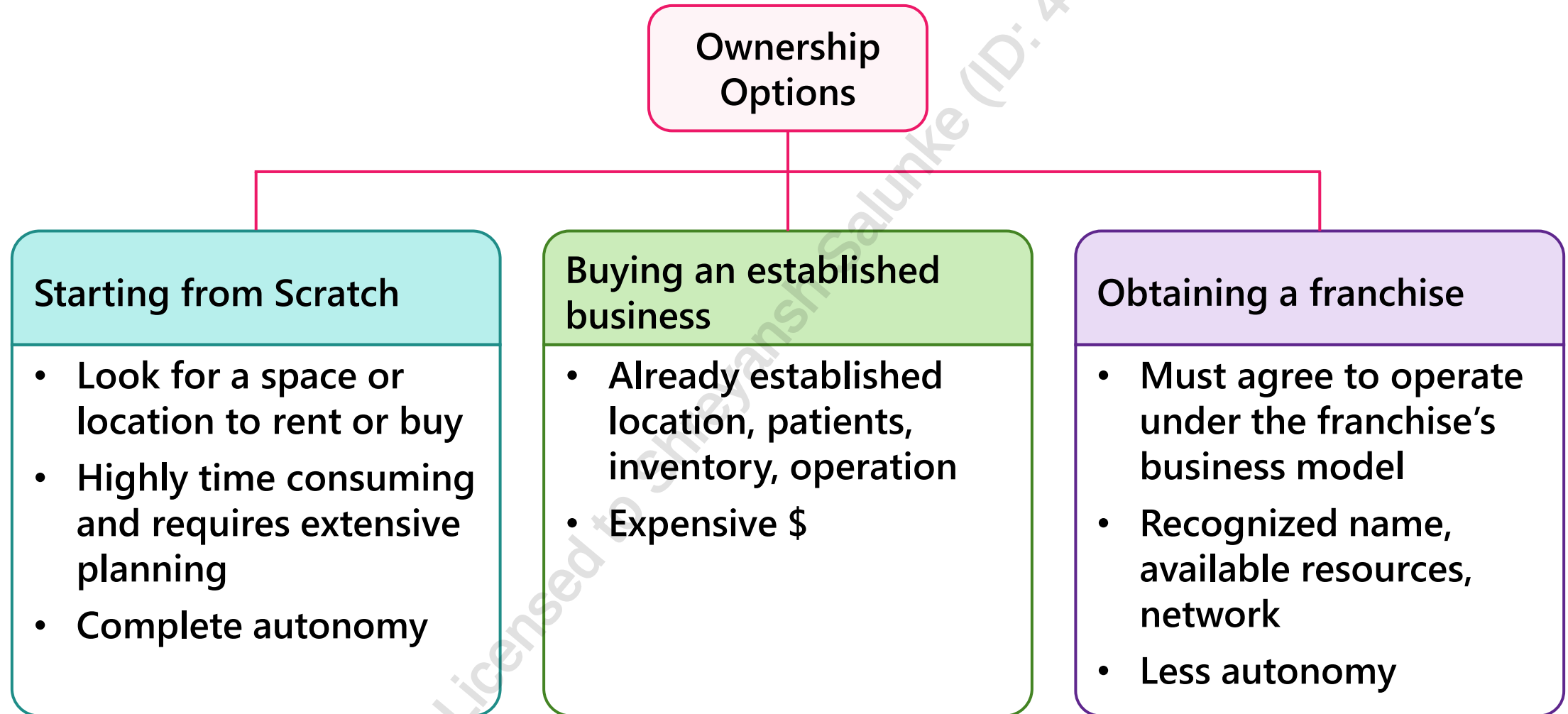
- A central fill pharmacy is a newer kind of pharmacy that has been created to help **support community pharmacies** and the role that they play in the community
- With expanded scopes of practice, prescription volume may be increasing. This creates an increasing workload on community pharmacies and central filling pharmacies offer a solution
- Community pharmacies can pay a portion of each prescription filled to a central fill pharmacy
- The central fill pharmacy will then fill the prescriptions for the community pharmacies

Kiosk

- **Self-serve pharmacy** where customers can speak with a pharmacist using videoconferencing technology that is embedded in the kiosk
- This kind of pharmacy is like a vending machine but for medications
- These are in downtown areas and some rural areas that are not serviced by other kinds of pharmacies

BUSINESS OPTIONS

FYI



BUSINESS ENTERPRISES

FYI

Type	Advantages	Disadvantages
Sole Proprietorship • Sole ownership • Business is legally indistinguishable from its owner	• Low startup costs • Owner has direct control • Greatest freedom from regulation • All profits go to the owner	• Unlimited liability • Difficult to raise capital • Lack of continuity • Personal and assets can be seized & sold • Difficult to keep best employees
Partnership General partnership • A business in which 2 or more people form a contractual relationship to run a business	• Low startup costs • More partners provide additional capitals • Taxed at personal rates • Easy to form	• Unlimited liability for each partner • Difficult to find suitable partner • Does not have separate identity (not distinguishable from partners)

BUSINESS ENTERPRISES

FYI

Type	Advantages	Disadvantages
Partnership Limited partnership • 1 or more partners has specific limitation on amount of partnership & debt for which she/he is liable for	• Limited partner not liable for debts beyond amount of capital contribution	• Limited partners do not have control over management • Issues include terms of partnership, sharing of earnings, management responsibilities, retirement/exit provisions
Franchise • Grants the rights (license) to operate under an established company	• Access to business' proprietary knowledge and operating methods • Use the above to run the business successfully	• Fee to be part of a franchise Not a form of ownership but an ownership style

BUSINESS ENTERPRISES

FYI

Type	Advantages	Disadvantages
Corporation • A legal business entity with authority to act & have liability separate from its owners	• Legal entity so owners are not personally liable • Easier to raise larger sum of capital • Easy to transfer ownership • Tax advantages via dividends • Ease of transfer of ownership	• Cost of incorporating business • Legal requirements of organizing & reporting extensive record keeping • Operating restrictions

Type of corporations

1. Small business: Corporation without share capital, private business, co-operative
2. Big business: Public company versus crown corporation
 - Public company: Shares may be available to public & listed for public trading on securities exchanges. Unlimited # of stakeholders. Usually 1 vote/share. No restrictions on share transfers.
 - Crown corporation: Partially or entirely owned by government (i.e. public transportation)

KEY STEPS IN OPENING A PHARMACY

FYI

1. Consult Advisors and Mentors
2. Research and create a business plan
3. Legal and operational requirements
4. Obtain a business bank account and startup capital
5. Select and Design a store
6. Hire and train employees
7. Purchase inventory and supplies
8. Marketing
9. Soft Opening
10. Grand Opening



MEDICATION SUPPLY CHAIN IN CANADA

Drug manufacturer submit research on a new drug product to Health Canada



Drug receives approval upon review of safety, efficacy and quality of the new drug product



Federal, provincial and/or territorial agencies managing drug prices will conduct a review to decide if the new medication will be included into the public drug plans and under what terms



Once procured, the pharmaceutical company can use distributors such as McKesson Canada and the Canadian Pharmaceutical Distribution Network to distribute their medications in bulk to pharmacies



Healthcare professionals prescribe, deliver and/or administer these medications to patients

PRODUCT ACQUISITION

- **Aim:** Stable, rapid and secure product delivery
- **Barriers:** backorders, recalls, counterfeit drugs, cost
- Acquired independently or through **buying groups**



McKesson CANADA

PRODUCT ACQUISITION

PURCHASING

Group purchasing organization (GPO)

- Pharmacies join a GPO, and the GPO negotiates with wholesaler on behalf of their members to provide the best rates
- Group decides on products to be tendered after which tenders are sent to suppliers
- Sealed bids are received by the group and the group reviews bid and decides on supplier based on price and number of criteria
- Common within hospitals

Pharmacy buying groups

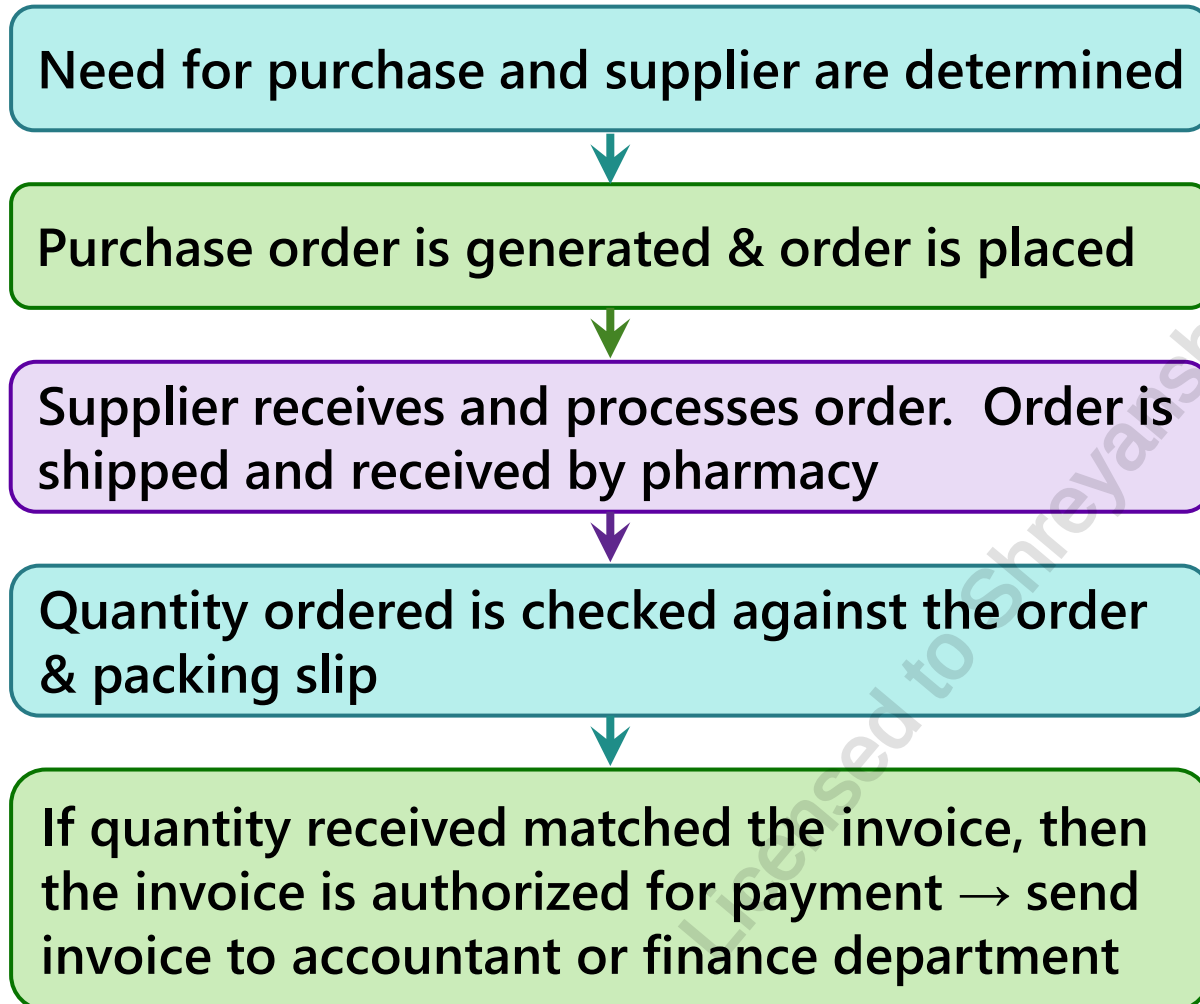
- Multiple pharmacies come together to form a group and negotiate a lower price with wholesalers
- Pharmacies charged an annual fee to be a part of buying group
- Used in large chains, banners (includes Rx, OTC, other front shop items)

Contract purchasing

- Purchasers agree to buy specific volume over a given time and at fixed price. Price reductions based on volumes
- Used in large chains, hospitals, banners (includes Rx, OTC, other front shop items)

PRODUCT ACQUISITION

PURCHASING PROCESS



PRODUCT ACQUISITION

PURCHASING DISCOUNTS

Quantity discounts

- The higher the quantity ordered at once, the lower the price (bulk purchasing)
- Can be at one time (12+ bottles in an order will give a 2% discount) or over a period of time (100 bottles in 4 months will give a 3% discount)

Cash discounts

- Common for suppliers to offer % discount for prompt payment
- E.g. 2% discount if payment is made before the scheduled due date

Serial discounts

- When multiple discounts are applied at the same time

PRODUCT ACQUISITION

PURCHASING STRATEGIES

Price Discrimination

- Different groups (e.g., students, seniors, etc.) are each charged different amounts for a service according to their ability to pay

Cost-Plus Pricing

- Prices calculated by taking the total cost of the product (e.g., cost of product including shipping + labor costs) and then adding a fixed percent (e.g., 20%)
- Ensures that the product is profitable

Purchasing strategies

Value Pricing

- Customers charged what they are willing to pay, as opposed to what the product costs
- Often used for unique products with little to no competition, such as certain patented drugs

Loss Leader

- Strategy where products/services are sold below cost to attract customers
- Customers who purchase these unprofitable products/services will also buy other products that are profitable to more than make-up for the loss

STORAGE

NATIONAL DRUG SCHEDULES (NDS)

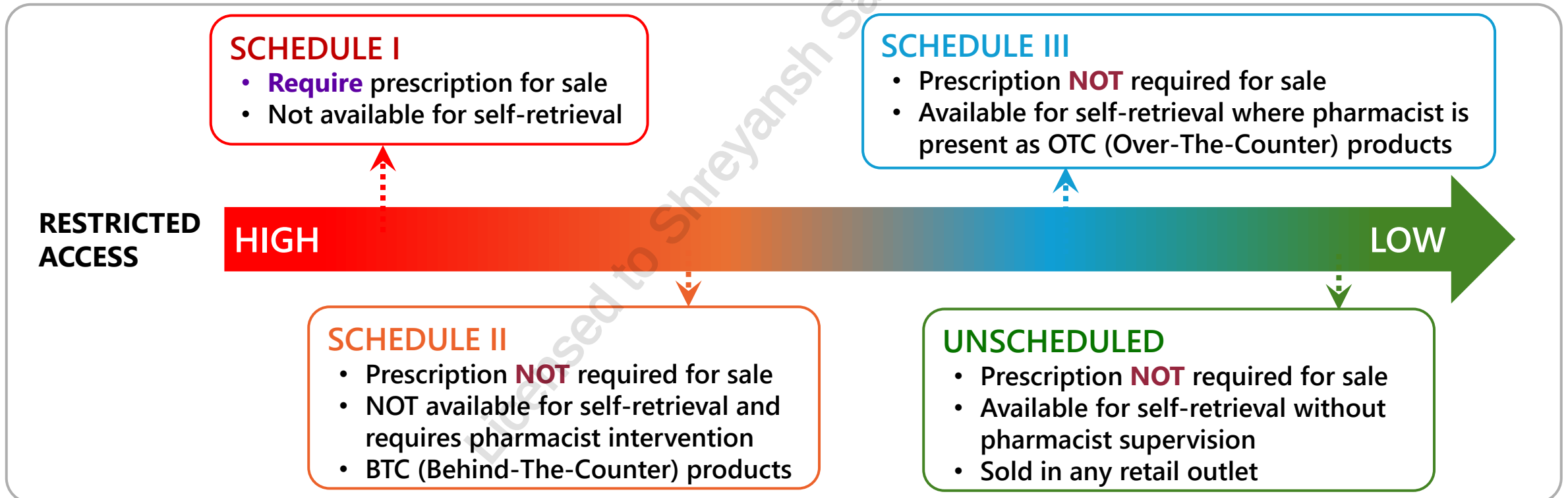
Composed of three schedules or four categories:

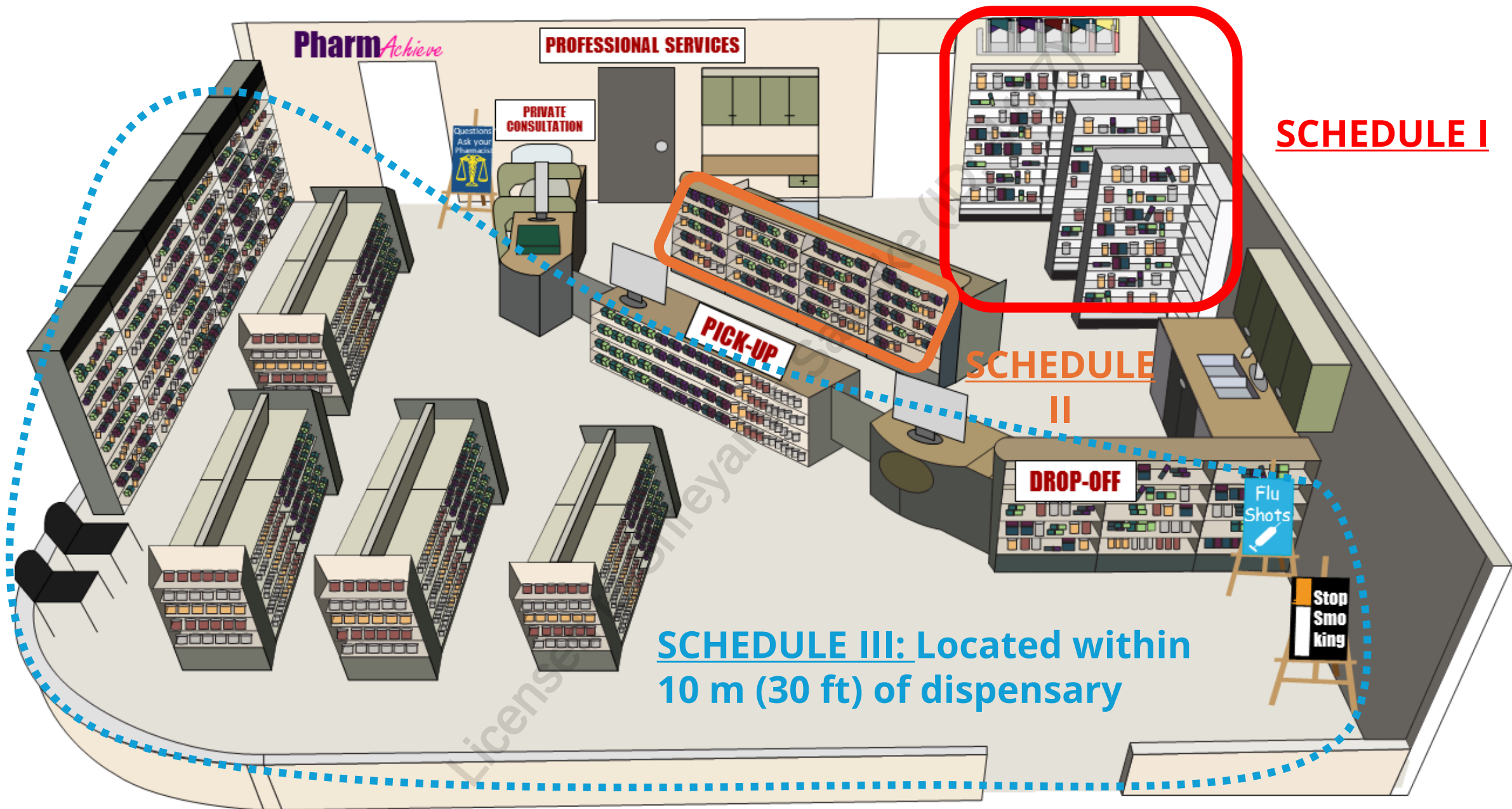
SCHEDULE I, SCHEDULE II, SCHEDULE III, UNSCHEDULED

- Establishes the conditions for sale of products which ultimately guides of the level of professional intervention necessary for the safe and effective use of drugs

NAPRA

National Association of
Pharmacy Regulatory Authorities





DRUG SCHEDULES FILL-IN-THE-CHART EXERCISE

Schedule	I	II	III	U
Do you need an Rx?				
Available for self retrieval?				
RPh Intervention required?				
Location				
Examples				

NARCOTIC STORAGE





1. What is the difference between each product?
2. What is the drug schedule of each?
3. Where should each product be stored in the pharmacy (think in terms of public access and visibility)?

STORAGE

REFRIGERATION



WHAT IS A COLD CHAIN?

"Cold chain is a temperature-controlled supply chain for products such as pharmaceuticals, vaccines and blood products that require a specific temperature range during distribution and storage" - Ontario College of Pharmacists (OCP)

- **Vaccines** usually stored between **2 °C and 8 °C**, with an alarmed thermometer
- **Frozen vaccines** should be kept at **-15 °C or colder** (depending on the product)
- Cold chain storage for vaccines involves steps taken to ensure optimal environments from the time of manufacture to the time of administration to the patient
 - Keep separate from other non-drug items (e.g. food)
 - Vaccines should be stored in the middle of the fridge, do not store vaccines on door shelves
 - Use plastic water bottles on door shelves and lower compartments to help stabilize temperature
 - Ensure the fridge door is always closed and is not unnecessarily opened
 - Avoid having vaccines in direct contact with ice packs if transporting out of the pharmacy (use insulated storage container)

STORAGE

REFRIGERATION



In the event of a break in cold chain:

- A break in the cold chain occurs when the storage of vaccines falls outside the recommended temperature range
 - Can result in the vaccine losing potency and lower protection against disease

Steps to take if a break in cold chain occurs:

1. Separate the affected supply and label "DO NOT USE"
 - If there is a failure in the fridge unit or a power outage, move the vaccines to an alternative location (i.e., cooler or storage unit)
2. Notify the delegate in charge of cold chain management
3. Record the incident
4. Contact local public health or the manufacturer to determine if the vaccine is still viable
5. If not useable, discard the vaccine

STORAGE

REFRIGERATION

Product	Comments
Insulin	Store in fridge until first use, then store at room temperature for 28 days. Do not use if frozen. Once opened, insulin degludec FlexTouch® can be stored at room temperature or in fridge for up to 8 weeks. Insulin Degludec Penfill® should be stored at room temperature for 8 weeks once opened
Monoclonal Antibodies	Store in fridge
GLP agonists	Semaglutide (Ozempic®, Wegovy® multi-use pen): Store in fridge until first use, then store in fridge or at room temperature for 8 weeks Liraglutide (Victoza®): Store in fridge until first use, then store in fridge or at room temperature for 30 days Dulaglutide (Trulicity®): Store in fridge until first use, then store in fridge or at room temperature for 14 days Tirzepatide (Mounjaro®) KwikPen: Store in fridge until first use, then store in fridge or room temperature for up to 30 days
Eye drops	Latanoprost Multidose bottle (Xalatan®): Store in fridge until opened, then stable for 6 weeks at room temperature Latanoprostene (Vyzulta®): Store in fridge until opened, then stable at room temperature for 8 weeks

STORAGE

REFRIGERATION

Product	Comments
Clavulin® -125 suspension	Store for 10 days in fridge once reconstituted
Clavulin® -200 suspension	Store for 7 days in fridge once reconstituted
Clavulin® -250 suspension	Store for 10 days in fridge once reconstituted
Clavulin® -400 suspension	Store for 7 days in fridge once reconstituted
Amoxicillin suspension	Can be stable at room temperature for 7 days after reconstitution, but up to 14 days if refrigerated
Azithromycin suspension	Store for 10 days in fridge or at room temperature once reconstituted
Cephalexin suspension	Store for 14 days in fridge once reconstituted
Cloxacillin suspension	Store for 14 days in fridge once reconstituted



***Reconstituted clindamycin and clarithromycin oral suspensions should not be stored in the fridge and are stable at room temperature for 14 days**

MONOGRAPHS

changes, stop using XALATAN and contact your doctor immediately.

XALATAN may cause the following side effects as well.

Common Ocular Side Effects: burning and stinging, blurred vision, red eyes, foreign body sensation, itching, increased iris pigmentation, damage of the cornea in a pinpoint pattern, dry eyes, excessive tearing, eye pain, eye lid crusting, red and swollen eyelid, eyelid discomfort/ pain, photophobia (visual sensitivity to light)

Uncommon Ocular Side Effects: discharge from the eye, diplopia (doubled vision), conjunctivitis, iritis/uveitis (inflammation of the interior of the eye), darkening of the palpebral skin (skin related to the eyelid)

Common Systemic Side Effects: upper respiratory tract infection/ cold/ flu, pain in muscle/ joint/ back, chest pain/angina pectoris, rash/ allergic skin reaction

This is not a complete list of side effects. For any unexpected effects while taking XALATAN, contact your doctor or pharmacist.

HOW TO STORE IT

Before XALATAN is first opened, keep it in a fridge (between 2°C and 8°C/36°F and 46°F), out of direct light. Once the bottle has been opened, XALATAN may be kept at room temperature up to 25°C. XALATAN must be used within 6 weeks after opening the bottle. Discard the bottle and/or unused contents after 6 weeks. XALATAN should not be used after the expiry date on the bottle.

Keep all medicines in a safe place, out of the reach of children.

ANTIBIOTIC STORAGE EXERCISE

	Refrigerate	Room Temperature	Either
Amoxicillin			
Amoxi-Clav			
Azithromycin			
Cloxacillin			
Cephalexin			
Clarithromycin			
Clindamycin			

ANTIBIOTIC STORAGE EXERCISE

	Refrigerate	Room Temperature	Either
Amoxicillin	✓	✓	✓*
Amoxi-Clav	✓		
Azithromycin	✓	✓	✓
Cloxacillin	✓		
Cephalexin	✓		
Clarithromycin		✓	
Clindamycin		✓	

*Refrigeration preferred

CASE

Background: A mother comes to your pharmacy and brings a prescription for her child, Lily, who weighs 18 kg. The prescription is for amoxicillin/clavulanate (liquid suspension) with a dose of 75 mg/kg/day of amoxicillin, divided TID for 10 days.

The brand of amoxicillin/clavulanate you have available in your pharmacy is Clavulin-400® (containing 400 mg amoxicillin and 57 mg clavulanate per 5 mL), which is available in a bottle size of 70 mLs.

How many mLs of Clavulin® does Lily have to take TID?

How many reconstituted bottles do we have to give Lily?



STORAGE

EXPIRED STOCK

- Inventory is checked regularly to ensure it is not expired
- Drugs close to expiry date should be clearly labeled and used first
- **Medication that will expire while the patient is still taking it should not be dispensed**
 - If a medication expires in 3 months and the patient is receiving a 3-month supply, the expiring medication should not be dispensed
 - If a medication expires in 3 months and the patient is receiving a 1-month supply, the expiring medication may be dispensed with clear warning to use before the expiry date
- In some cases, unopened bottles of expired stock can be sent back to the manufacturer and the pharmacy may receive some money back for it
- If the expired stock cannot be returned to the manufacturer, it should be safely disposed of in the pharmacy disposal bins

STORAGE

SHARPS DISPOSAL

Sharps, such as used needles, should not be placed in the garbage, they should be **appropriately disposed of in a sharps container**

- Safe disposal helps to reduce the risk of needlestick injuries and harm to the environment
- Sharps containers can be found at most pharmacies and are free of charge for patients to obtain
- Pharmacy staff are advised to provide patients with one whenever dispensing injectable products, and instruct patients to return the container once full
- The sharps container allows for the **one-way disposal of sharps**, and cannot be opened to retrieve needles once deposited



PHARMACY STAFF ROLES

PHARMACIST (RPh)

- Can perform all pharmacy tasks
- Usually performs mostly clinical tasks

	REGISTERED PHARMACY TECHNICIAN (RPhT)	REGISTERED PHARMACY STUDENT/INTERN	PHARMACY ASSISTANT
Receive new and repeat verbal prescriptions from other HCPs	 Not controlled substances		
Perform independent technical prescription checks			
Accepting/sending transfer prescriptions from other pharmacies	 Not benzodiazepines and targeted substances		
Perform test on tissue below dermis			
Administer vaccine	 Only publicly funded		
Prescribe (Minor Ailment, Smoking Cessation, Renewing or Adapting Existing Rx)			

DISPENSING PROCESS

PRESCRIBING



Prescribing

- Prescription written by HCP Practitioner (e.g. Physician, Veterinarian, Dentist) or New Class of Practitioners (e.g. Nurse Practitioner, Midwife, Podiatrist) or Pharmacist with prescribing authority (in provinces where this exists)
- Rx **can** be hand-written, printed, faxed, e-prescribed
- Rx **cannot** be emailed (since e-mail is considered insecure)
- May be partially pre-printed – What does that mean?

Can a physician's secretary verbally (by mouth) relay prescription information to the pharmacist?

DISPENSING PROCESS

PRESCRIBING



Verification of Rx Authenticity

- What are the requirements for a new prescription?
 - Patient's name, government issued identification # (for monitored substances), prescriber's contact information (phone number, address, name, license number), prescriber signature, drug to be dispensed (name, strength) directions for use (frequency, dose, quantity), refills* or part-fills, date Rx was written
 - For verbal prescription: Name of recording pharmacy staff member and license number
- Faxes must be directly from prescribers
- Must contain an image of a **signature** for verification of authenticity
- Must also include the **date**
- Be suspicious of large quantities in unfamiliar patients
- Call practitioner when in doubt

*No refills for narcotic drugs



DISPENSING PROCESS

PRESCRIBING

Can this prescription be filled as written?

Eva's health card number is missing.
A government-issued ID number must
be on every narcotic prescription.

Naturopathic Doctors (NDs), are not
authorized to prescribe narcotics.
Medical Doctors (MDs), Nurse
Practitioners (NPs), midwives, dentists,
vets and podiatrists are authorized to
prescribe narcotics.

MEDICAL CENTRE
ADDRESS
PHONE NUMBER

Today's Date

Eva Liu
34 Woodstock Drive
London ON
DOB: March 12 1975
Tel# 225-239-2019

Oxycodone 20 mg IR tabs
T1T PO Q6H PRN
Mitte: 100 tabs
Pharmacist instructions: Dispense 30 tabs every 5 days
Repeats: 1


Dr. Marie Allan, ND #3346

Refills are not
permitted on
narcotic
prescriptions.

DISPENSING PROCESS

PRESCRIBING- FRADULANT PRESCRIPTIONS

1. Assess the patient

- Do you know the patient?
- Is the medication appropriate and consistent with the treatment history?
- Is the patient uncomfortable or nervous? Avoiding eye contact?
- Is the patient intimidating the staff, showing aggression or is in a hurry?

2. Evaluate the prescription

- Do you know the prescriber? Does the information on public register of the prescriber regulatory college confirm the information on the prescription?
- Is the Prescriber out of town?
- Examine the prescription through Narcotic Monitoring System (NMS) if Controlled substance
- Is the quantity reasonable?
- Is the prescription a photocopy and not original?

DISPENSING PROCESS

PRESCRIBING- COMMON ERRORS

Verbal Prescription

- Prescriber's tone / accent / speed
- Environment (e.g. busy, noisy)
- Number clarity (15 vs 50)
- Sound-alike medications (e.g. dimenhydrinate vs diphenhydramine)
- Similar/same patient name
- Closed-ended questions

Written Prescription

- Illegible handwriting
- Abbreviations (e.g. OD and od)
- Look-alike medications (e.g. Toradol® and tramadol)
- Similar patient information
- Poor fax copy/transmission

DO NOT USE

ISMP'S DANGEROUS ABBREVIATIONS, SYMBOLS, AND DOSE DESIGNATIONS

Abbreviation	Intended Meaning	Problem	Correction
U	unit	Mistaken for "O" (zero), "4" (four), or cc	Use "unit"
IU	international unit	Mistaken for "IV" (intravenous) or "10" (ten)	Use "unit"
Abbreviations for drug names		Misinterpreted because of similar abbreviations for multiple drugs; e.g., MS, MSO ₄ (morphine sulphate), MgSO ₄ (magnesium sulphate) may be confused for one another	Do not abbreviate drug names
QD QOD	Every day Every other day	QD and QOD have been mistaken for each other, or as 'qid'. The Q has also been misinterpreted as "2" (two)	Use "daily" and "every other day"
OD	Every day	Mistaken for "right eye" (OD = oculus dexter)	Use "daily"
OS, OD, OU	left eye, right eye, both eyes	May be confused with one another	Use "left eye", "right eye" or "both eyes"
D/C	Discharge	Interpreted as "discontinue whatever medications follow" (typically discharge medications)	Use "discharge"
cc	cubic centimetre	Mistaken for "u" (units)	Use "mL" or "millilitre"
µg	microgram	Mistaken for "mg" (milligram) resulting in one thousand-fold overdose	Use "mcg"

DO NOT USE

ISMP'S DANGEROUS ABBREVIATIONS, SYMBOLS, AND DOSE DESIGNATIONS

Symbol	Intended Meaning	Potential Problem	Correction
@	at	Mistaken for "2" (two) or "5" (five)	Use "at"
> <	Greater than Less than	Mistaken for "7"(seven) or the letter "L". Confused with each other	Use "greater than"/"more than" or "less than"/"lower than"

Dose Designation	Intended Meaning	Potential Problem	Correction
Trailing zero	$x.0$ mg	Decimal point is overlooked resulting in 10-fold dose error	Never use a zero by itself after a decimal point. Use " x mg"
Lack of leading zero	$.x$ mg	Decimal point is overlooked resulting in 10-fold dose error	Always use a zero before a decimal point. Use " $0.x$ mg"

DISPENSING PROCESS

TRANSCRIBING



Transcribing: Order Entry

- Details of prescription are entered into the pharmacy system e.g. Nexxsys, Kroll, HealthWatch
- Any pharmacy team member can transcribe

Interchangeability

- Same quantity of active ingredients, comparable pharmacokinetics
- Same route of administration & dosing schedule
- Price does NOT exceed that of "brand"
- When should the brand name be used?

If a new prescription is faxed over to the pharmacy for a new patient, would you fill brand or generic (assume patient is not answering the phone and is expecting the medication to be ready in 15 minutes)?

DISPENSING PROCESS

TRANSCRIBING- BRAND, GENERIC, BIOLOGIC, BIOSIMILAR

BRAND NAME

- Known as the innovator or reference drug
- Marketed as new chemical entities
- More expensive than generic

BIOLOGIC

- A biologic medicine is derived from living organisms and is manufactured through highly involved and stringently controlled biotechnology processes.
- As biologic medicines, biosimilars are inherently different from generics due to their molecular size and structure, and the complexity and cost of their development

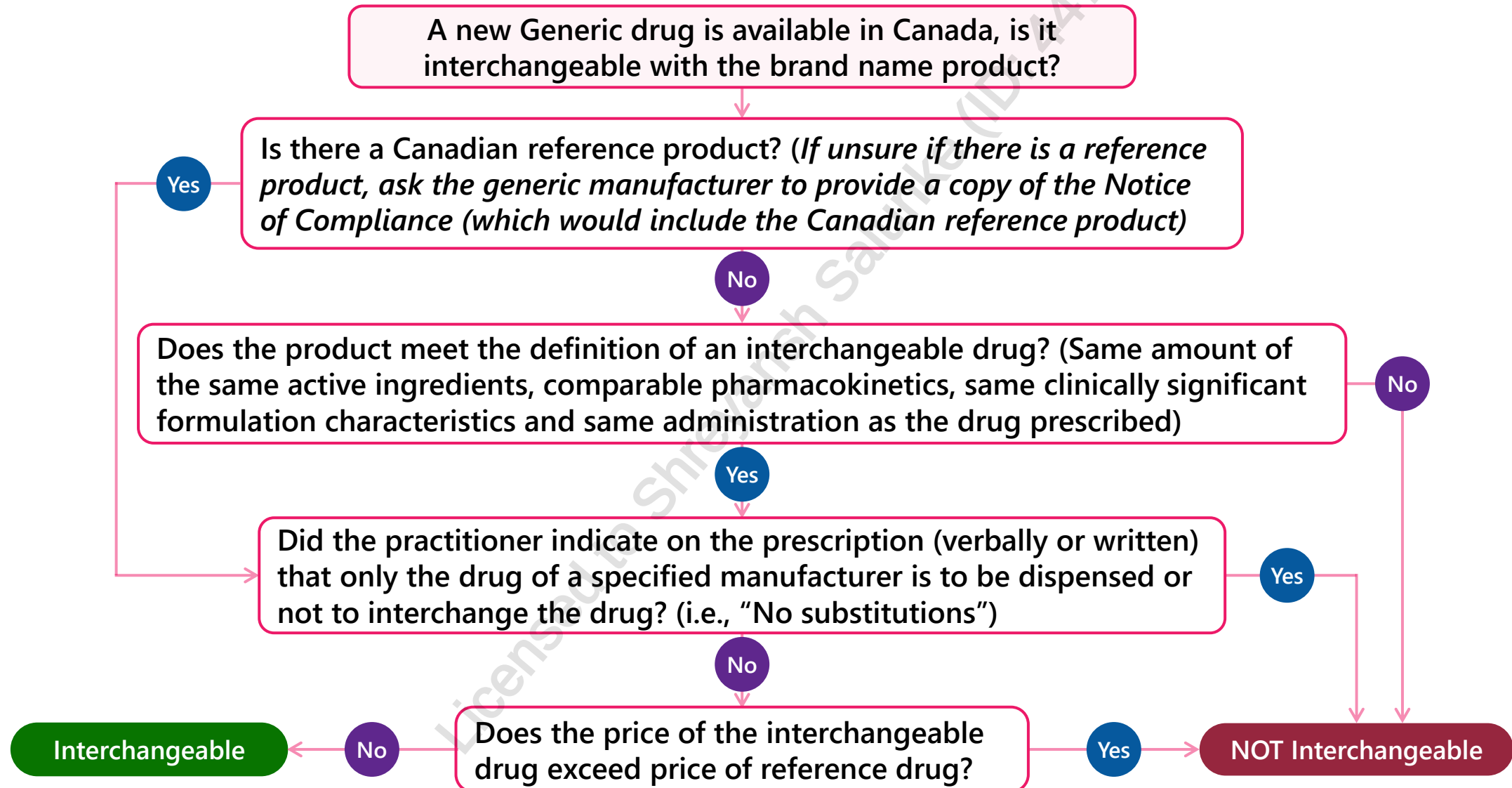
GENERIC

- Generic drugs are small molecules that are chemically synthesized
- They contain **identical** medicinal (active) ingredients to their reference products and meet bioequivalence standards by Health Canada
- Produced after the original patent expires

BIOSIMILAR

- A biosimilar is a drug that is highly similar but **NOT identical** to a biologic drug that was already authorized for sale → NOT automatically interchangeable
- Health Canada authorizes biosimilars for sale using the same rigorous regulatory standards for quality, efficacy and safety as for all other biologic drugs

DISPENSING PROCESS – TRANSCRIBING - INTERCHANGEABILITY



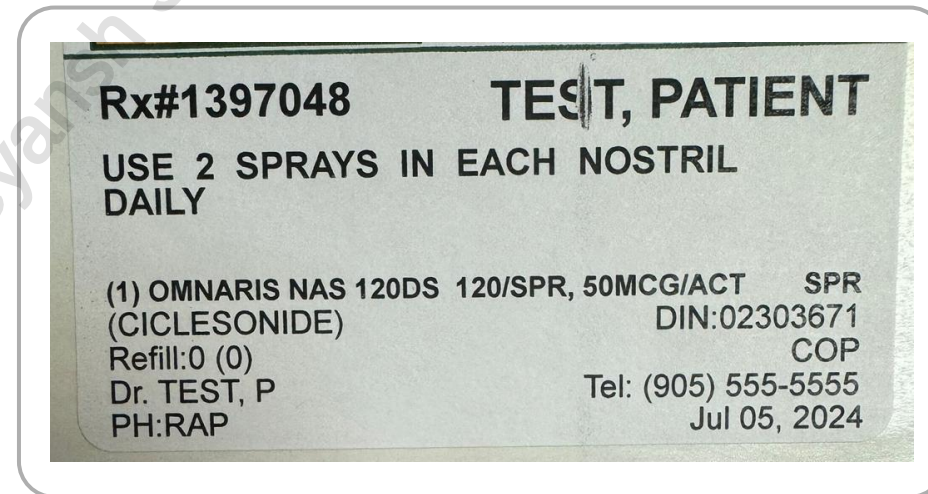
DISPENSING PROCESS

PACKAGING



PACKAGING

- Also known as “filling” - performed by RPhT, assistant, student, intern and/or RPh
- Ensure the patient label is correct
- Select the correct drug, dose, brand, quantity and dosage form
- Check expiry
- Apply appropriate auxiliary labels (e.g. take with food, store in fridge)



DISPENSING PROCESS

PACKAGING- DISPENSING CHECKS

TECHNICAL CHECKS

- Can be performed by RPhT, intern, pharmacy student
- What does a technical check entail?
 - *Correct patient, drug dose/form/route, doctor, quantity, number of refills*
- Can any pharmacy staff prepare blister packs?
 - *Yes, however the final check must be completed by a registered technician (if unavailable, the pharmacist may perform final check as well)*
- What is an “independent double check?”
 - *Involves a verification process performed by two authorized individuals without influence by one another's results*

CLINICAL CHECKS

- Only performed by RPh, intern, pharmacy student
- What does a clinical check entail?
 - *Therapeutic appropriateness and consultation/counselling*
- Can pharmacy students/interns perform clinical checks independently?
 - *Yes!*

DISPENSING PROCESS

ADMINISTRATION- PATIENT COUNSELLING



- Completed by the pharmacist, intern, or student
- Registered Pharmacy Technician can provide information and education if it isn't clinical (e.g. can provide information on how to use a blood glucose machine but cannot interpret the meaning of the numbers for the patient)
- Confirm the patient identity
- Review the medication name(s) and indication(s)
- Describe the dose, route and frequency
- Discuss important clinical information (adverse drug reactions, time to effect, interactions, etc.)
- Have the patient summarize and/or demonstrate vital information to ensure understanding



DISPENSING PROCESS

MONITORING



- Schedule follow-up appointments (often by phone) to monitor the patient's progress
- Adjust medication as needed based on patient response
- Report any adverse reactions or issues to the prescribing physician
- Especially important for minor ailments

DISPENSING PROCESS

MONITORING - RECALL

Recall: the decision by a pharmaceutical company to remove products from consumer access due to health risk

Three types of recall

1. Type I: Use of product will cause serious health consequences or death
2. Type II: Use of product may cause temporary adverse health effects
3. Type III: Use of product is unlikely to cause harm

Recall depth: process to determine at which level in the supply chain a recall is extended

1. Wholesale level: products are removed before reaching retailer or consumer
2. Retail level: affected products are removed from shelves
3. Consumer level: individuals who may be affected by recalled product

PRACTICE QUESTIONS

1. A physician is calling in prescriptions for venlafaxine and clonazepam. A Registered pharmacy Technician (RPhT) answers the phone at your community pharmacy. Can the RPhT accept both prescriptions?
2. A patient comes into your pharmacy asking for help on how to use their blood pressure monitor. Can the RPhT perform this task?
 - Follow up question to #2: Can the RPhT provide counselling on what the numbers mean? Why or why not?

PRACTICE QUESTIONS

3. A new patient comes into the pharmacy presenting a prescription for Teva-Lenoltec No.3. Upon viewing the prescription, you notice the patient identification number is missing. What is the most appropriate course of action?
- a) Dispense the medication as written
 - b) Contact the prescriber
 - c) Hold the medication and consider it a forgery
 - d) Send patient to another pharmacy that can fill the prescription
4. A patient comes into the pharmacy asking to check their fasting blood glucose. They ask the pharmacy assistant to assist in performing the glucose check. Can the assistant help this patient? Why or why not?

TAKEAWAY

- Common types of pharmacy formats: chain, banner, independent, franchise and central fill
- National Drug Schedules: Schedule I (prescription), II (behind the counter), III (over the counter) and unscheduled (found anywhere)
- Brand name medications are known as the reference drug and are new chemical entities. Generic medications contain the same active ingredient as a brand and meet the bioequivalence standard.
 - These two forms can be considered **interchangeable** as they are considered identical
- Biologics and biosimilars are “highly similar” drugs, however, are not identical. Therefore, these products are not interchangeable
- Stock is checked **every quarter** to ensure it is not expired
- Vaccines usually stored between **2 °C and 8 °C**, with an alarmed thermometer
- **Dispensing process**: prescribing, transcribing, packaging, counselling, monitoring
- 3 types of recalls: Type 1 (risk of severe consequences or death), Type 2 (temporary effects), and Type 3 (unlikely to cause harm)

REFERENCES

1. OCP Website, Accessed On January 18, 2024, <https://www.ocpinfo.com/Practice-education/Opening-operating-pharmacy/Frequently-asked-questions/>
2. How To Open Own Pharmacy, Accessed On January 18, 2024, <https://www.pioneerrx.com/how-to-open-a-pharmacy>
3. NAPRA Drug Schedules, Accessed On January 25, 2024, <https://napra.ca/Pharmacy-regulation-canada>
4. Protecting The Cold Chain, Accessed On February 8, 2024 https://www.ocpinfo.com/Wp-content/uploads/Documents/Protecting_the_cold_chain_part_i.pdf
5. CHAPTER 1 - PHARMACY IN CANADA, Accessed On January 18, 2024 https://northrose.ca/Previews/CPC_01.pdf
6. Medication Supply Chain, Accessed On January 25, 2024 <https://badgut.org/information-centre/a-z-digestive-topics/medication-supply-chain/>
7. Monograph Consulted For Each Drug Product On Slide 21
8. Identifying Forgeries Accessed On February 8, 2024, <https://www.ocpinfo.com/Practice-education/Practice-tools/Fact-sheets/Forgeries/>
9. Biologic, Biosimilar, Generic, Brand Accessed On February 8, 2024, <https://www.canada.ca/en/health-canada/services/drugs-health-products/biologics-radiopharmaceuticals-genetic-therapies/applications-submissions/guidance-documents/fact-sheet-biosimilars.html#:~:text=biosimilars%20are%20not%20the%20same,highly%20similar%2c%20but%20not%20identical.> Also, <https://www.pfizer.com/sites/default/files/investors/financial-reports/annual-reports/2018/our-innovation/progressing-our-science/building-upon-heritage/index.html>
10. Slide 46 Chart Adapted From Drug Interchangeability Update In *FYI Newsletter*, BC College Of Pharmacists, 2004
11. Pharmacy partnerships to grow purchasing Power | McKesson. (n.d.). <https://www.mckesson.com/pharmacy-management/group-purchasing/>
12. Brown, C., Deeter, B., Forest, C., Fortier, D., Smith, A. G., Van Haarlemm, L., Ilek-Priouzeau, S., Laroche, J., Oliver, G., Richardson, R., & Salinas, C. (2015). National Vaccine Storage and Handling Guidelines for Immunization Providers.
13. Public Health Agency of Canada. (2022, September 16). Storage and handling of immunizing agents: Canadian Immunization Guide. Canada.ca. <https://www.canada.ca/en/public-health/services/publications/healthy-living/canadian-immunization-guide-part-1-key-immunization-information/page-9-storage-handling-immunizing-agents.html>
14. Health Protection Branch & Health Canada. (1993). PRODUCT RECALL PROCEDURES. https://www.canada.ca/content/dam/hc-sc/migration/hc-sc/dhp-mps/alt_formats/hpfb-dgpsa/pdf/compli-conform/recall-proc-marche_retrait-eng.pdf

CHANGE LOG

August 2025

- Formatting and grammatical edits made throughout
- Slide 13: Added link to McKesson Canada
- Slide 14: Removed 'hospitals' from pharmacy buying groups
- Slide 16: Edited definitions of cash discounts and serial discounts
- Slide 22: Edited temperature for frozen vaccines
- Slide 24: Edited storage information of insulin Degludec
- Slide 47: Minor edits to Type II recall

October 2025

- Formatting and grammatical edits made throughout
- Slide 24: Addition of Wegovy® and Mounjaro®
- Slide 30: Reworded first bullet point
- Slide 34: Removal of 'Look for non-matching ink-Be familiar with handwriting of local physicians'

January 2026

- Slides 7-11: Labelled as FYI

February 2026

- Slides 20: Additional exercise for drug schedules
- Slide 35: Addition of note indicating no refills for narcotic drugs
- Slide 36: Included vets as authorized prescriber
- Slide 42: Addition of note indicating biosimilars are not interchangeable